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CS 621 DATA ANALYTICS

Web and Social Network Data

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UNSTRUCTURED TEXT

- Online news
- e-Books
- Web pages and blogs
- Social media contents – tweets, instagrams
- Product reviews – amazon/ebay/yelp, server logs (mixed types of data)

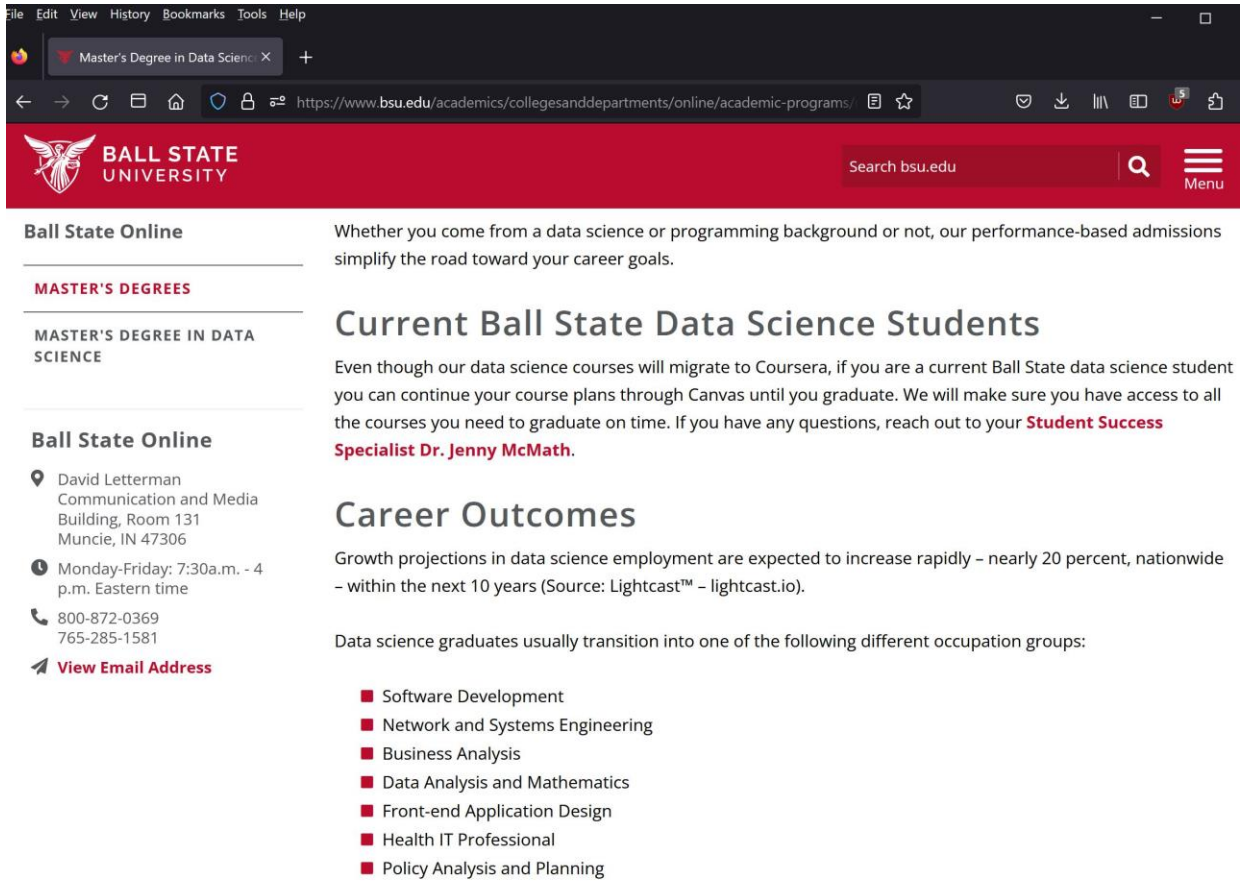
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WEB PAGE HTML

- CSS (Styles)
- Javascript (Dynamic effect)

- HTML structures (XML format)
  - html tags, contents, attributes
  - elements of interest:
    - headers, e.g, <h1></h1>
    -
    - <p>
    - <table><tr><td></td></tr></table>
    -

A BSU WEB PAGE



File Edit View History Bookmarks Tools Help

Master's Degree in Data Scienc

https://www.bsu.edu/academics/collegesanddepartments/online/academic-programs/

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Search bsu.edu

Menu

Ball State Online

MASTER'S DEGREES

MASTER'S DEGREE IN DATA SCIENCE

Ball State Online

David Letterman Communication and Media Building, Room 131
Muncie, IN 47306

Monday-Friday: 7:30a.m. - 4 p.m. Eastern time

800-872-0369
765-285-1581

View Email Address

Whether you come from a data science or programming background or not, our performance-based admissions simplify the road toward your career goals.

Current Ball State Data Science Students

Even though our data science courses will migrate to Coursera, if you are a current Ball State data science student you can continue your course plans through Canvas until you graduate. We will make sure you have access to all the courses you need to graduate on time. If you have any questions, reach out to your **Student Success Specialist Dr. Jenny McMath**.

Career Outcomes

Growth projections in data science employment are expected to increase rapidly – nearly 20 percent, nationwide – within the next 10 years (Source: Lightcast™ – lightcast.io).

Data science graduates usually transition into one of the following different occupation groups:

- Software Development
- Network and Systems Engineering
- Business Analysis
- Data Analysis and Mathematics
- Front-end Application Design
- Health IT Professional
- Policy Analysis and Planning

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WEB PAGE SOURCE

```
<div class='component rte'><h2>About the Master&rsquo;s in Data Science </h2>
<p>Data science is not just a popular career path, it&rsquo;s one that&rsquo;s here to stay. Employers are searching
<p>The <strong>33-credit, non-thesis program</strong> provides you with 15 credits of core courses and 18 credits to
<h2>Who Should Enroll </h2>
<p>The short answer is anyone! This program is built for learners of all educational backgrounds and career experienc
<h3>Performance-Based Admissions </h3>
<p>You can gain acceptance into the master&rsquo;s of data science through our performance-based admissions. No tests
<!-- End Component: RTE -->

<!-- Component: RTE -->



<div class='component rte'><p>Ball State University has partnered with Coursera, one of the largest online learning p
<p>Whether you come from a data science or programming background or not, our performance-based admissions simplify t
<!-- End Component: RTE -->
<!-- Start Accordion --><div class='component accordion'><div class='panel-group' id='accordion2E2A610E09E14FC09228E8

</div></div><!-- End Accordion -->

<!-- Component: RTE -->



<div class='component rte'><h2>Current Ball State Data Science Students </h2>
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<!-- Component: RTE -->



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<ul>
    <li>Software Development</li>
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    <li>Business Analysis</li>
    <li>Data Analysis and Mathematics</li>
    <li>Front-end Application Design</li>
    <li>Health IT Professional</li>
    <li>Policy Analysis and Planning</li>
</ul>
<ul>
```

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RETRIEVE WEB DATA

- from urllib.request import urlopen, Request
- req = Request(url)
- response = urlopen(req)
- print response.info()
- html = response.read()
- response.close()

- **Note**

  Because of internet security issues, many web servers prohibit automatic web scraping via software.

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BEAUTIFULSOUP

- from bs4 import BeautifulSoup
- soup = BeautifulSoup(html, “html.parser”)
- print(soup.prettify()) – show html structures

- print(soup.title)

- print(soup.title.string)

- print(soup.p)

- for p in soup.find_all(‘p’):
  print(p.text)

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TEXT DATA RETRIEVED

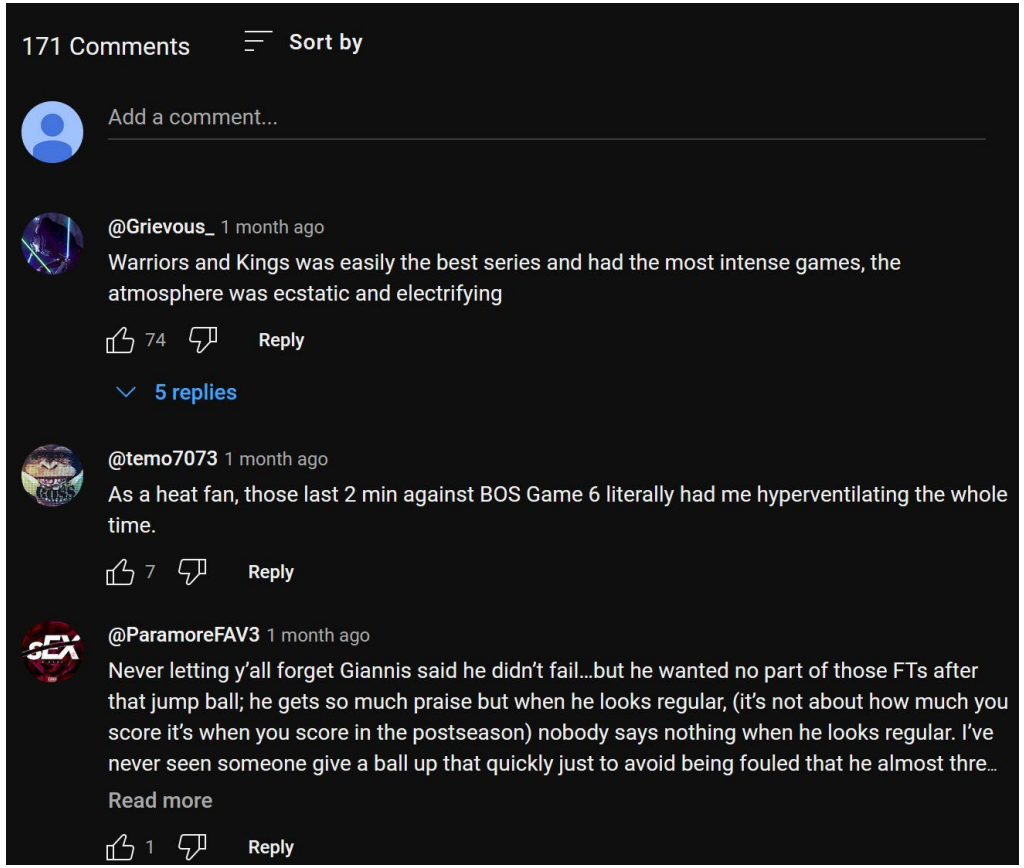
Ball State University has partnered with Coursera, one of the largest online learning platforms in the world, to bring you the online master of science in data science. The curriculum is designed by Ball State University, courses ar e taught by our accomplished faculty, and content is delivered through the Co ursera platform.
Whether you come from a data science or programming background or not, our pe rformance-based admissions simplify the road toward your career goals.
Data science is not just a popular career path, it’s one that’s here to stay. Employers are searching for those with the skills and knowledge to transform data to push their organizations forward. Ball State's online-only master's i n data science will help you gain these big data skills.
The 33-credit, non-thesis program provides you with 15 credits of core course s and 18 credits to focus your learning experience. The knowledge you gain in these courses can easily expand into several discipline-specific applications in the health, environmental, social, and behavioral sciences.
The short answer is anyone! This program is built for learners of all educati onal backgrounds and career experiences. Whether you’re working in an entry-l

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SOCIAL NETWORKS



Joe Biden now controls the @POTUS Twitter account. (Twitter)



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SOCIAL NETWORK DATA APIS

- Twitter API(now X)
  - https://developer.twitter.com/en/docs/twitter-api
- YouTube Data API
  - https://developers.google.com/youtube/v3
- Some may still have free data access after obtaining developer membership.

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TWEETS IN JSON

- Each returned tweet is a JSON object.
- Convert JSON strings to dictionaries.
- Access each object's value via its key value.
- The value of a key may contain more data in dictionary or list data structures. Thus, several rounds of parsing might be needed.

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KEYS IN TWEETS

- 'created_at', 'id', 'id_str', 'text', 'truncated', 'entities', 'source', 'in_reply_to_status_id', 'in_reply_to_status_id_str', 'in_reply_to_user_id', 'in_reply_to_user_id_str', 'in_reply_to_screen_name', 'user', 'geo', 'coordinates', 'place', 'contributors', 'is_quote_status', 'retweet_count', 'favorite_count', 'favorited', 'retweeted', 'possibly_sensitive', 'lang'

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A TWEET EXAMPLE


```
{'created_at': 'Thu Nov 03 11:01:44 +0000 2022',
 'id': 1588124261629116416,
 'id_str': '1588124261629116416',
 'text': 'Jo van Gogh Bonger was also known as ‘the woman who made Vincent famous’. Hans Luijten, Sen
ior Researcher at the Va… https://t.co/jf7ffyJnpO',
 'truncated': True,
 'entities': {'hashtags': [],
  'symbols': [],
  'user_mentions': [],
  'urls': [{'url': 'https://t.co/jf7ffyJnpO',
    'expanded_url': 'https://twitter.com/i/web/status/1588124261629116416',
    'display_url': 'twitter.com/i/web/status/1…',
    'indices': [117, 140]}]},
 'source': '<a href="https://www.hootsuite.com" rel="nofollow">Hootsuite Inc.</a>',
 'in_reply_to_status_id': None,
 'in_reply_to_status_id_str': None,
 'in_reply_to_user_id': None,
 'in_reply_to_user_id_str': None,
 'in_reply_to_screen_name': None,
 'user': {'id': 24691376,
  'id_str': '24691376',
  'name': 'Van Gogh Museum',
  'screen_name': 'vangoghmuseum',
  'location': 'Amsterdam',
  'description': 'Van Gogh Museum Amsterdam houses the world’s largest Vincent van Gogh collection: p
aintings, drawings, letters and more.',
  'url': 'https://t.co/a6IsctPqmQ',
  'entities': {'url': {'urls': [{'url': 'https://t.co/a6IsctPqmQ',
      'expanded_url': 'http://www.vangoghmuseum.com',
      'display_url': 'vangoghmuseum.com',
      'indices': [0, 23]}]},
   'description': {'urls': []}},
```

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TWEET CONTENTS RETRIEVED

```
0
Jo van Gogh Bonger was also known as 'the woman who made Vincent famous'. Hans Luijte

1
#QuizTime: Can you see what all of these portraits have in common? Let us know in the

2
We aim to make a positive impact on the lives of new generations around the world thr

3
Happy Halloween!💀 How are you celebrating today?

🎨 Vincent van Gogh, Skull (1887) https://t.co/XJUBLaTsK7

4
An acquaintance of Van Gogh's in Paris described how he bought old work shoes at a fl

5
For Vincent van Gogh, finding his calling was a long process. But he wasn't alone: hi

6
Can you see which painting is by Vincent van Gogh and which is by Gustav Klimt?
Klimt was inspired by Vincent's pai… https://t.co/ccgZ3iCpEV

7
This still life is Van Gogh's ode to modern French literature. Vincent was an avid re

8
'I can't live without love, without a woman. I wouldn't care a fig for life if there

9
🎨Vincent only became an artist at the age of 27. His parents were not pleased with h
```



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WE FLY